General Biology 1

First Semester

Supplementary Materials

Module 3

Light vs. Electron Microscope

Raymart F. Redita
Author

What I Know

Directions: Read each situation carefully and choose the best answer.

1. Which microscope generally has greater magnification?

   a. Light microscope
   b. Electron microscope
   c. Hand lens
   d. Magnifying glass

2. Which statement best describes a light microscope?
   a. It uses electrons to form images
   b. It is commonly used to observe living cells
   c. It can magnify objects up to one million times
   d. It requires a vacuum environment
3. Which statement best describes an electron microscope?
   a. It is inexpensive and easy to use
   b. It can observe living cells directly
   c. It provides very high-resolution images
   d. It uses sunlight to form images
4. A researcher wants to study the internal structure of mitochondria in detail. Which microscope would be most suitable?
   a. Light microscope
   b. Electron microscope
   c. Hand lens
   d. Simple microscope
5. A researcher is comparing two microscopes. Microscope A can observe living cells but has limited resolution. Microscope B produces extremely detailed images but requires non-living samples. Which conclusion is most accurate?
   a. Microscope A is an electron microscope
   b. Microscope B is a light microscope
   c. Microscope A is a light microscope, and Microscope B is an electron microscope
   d. Both microscopes are electron microscopes.

What I Need to Know

This supplementary module is designed to help you understand the difference between a **light and an electron microscope**, one of the most important foundations in biology. In this lesson, you will explore the advantages and limitations of each microscope in observing cell.

Lesson 3 Light vs Electron Microscope

A microscope is an essential scientific tool that allows us to see objects too small for the naked eye. The function is to magnify and reveal the fine details of cells and other tiny structures, helping us understand the microscopic world. In this lesson, you will learn about the structure and function of light and electron microscopes.

After going through this supplementary module, you are expected to:

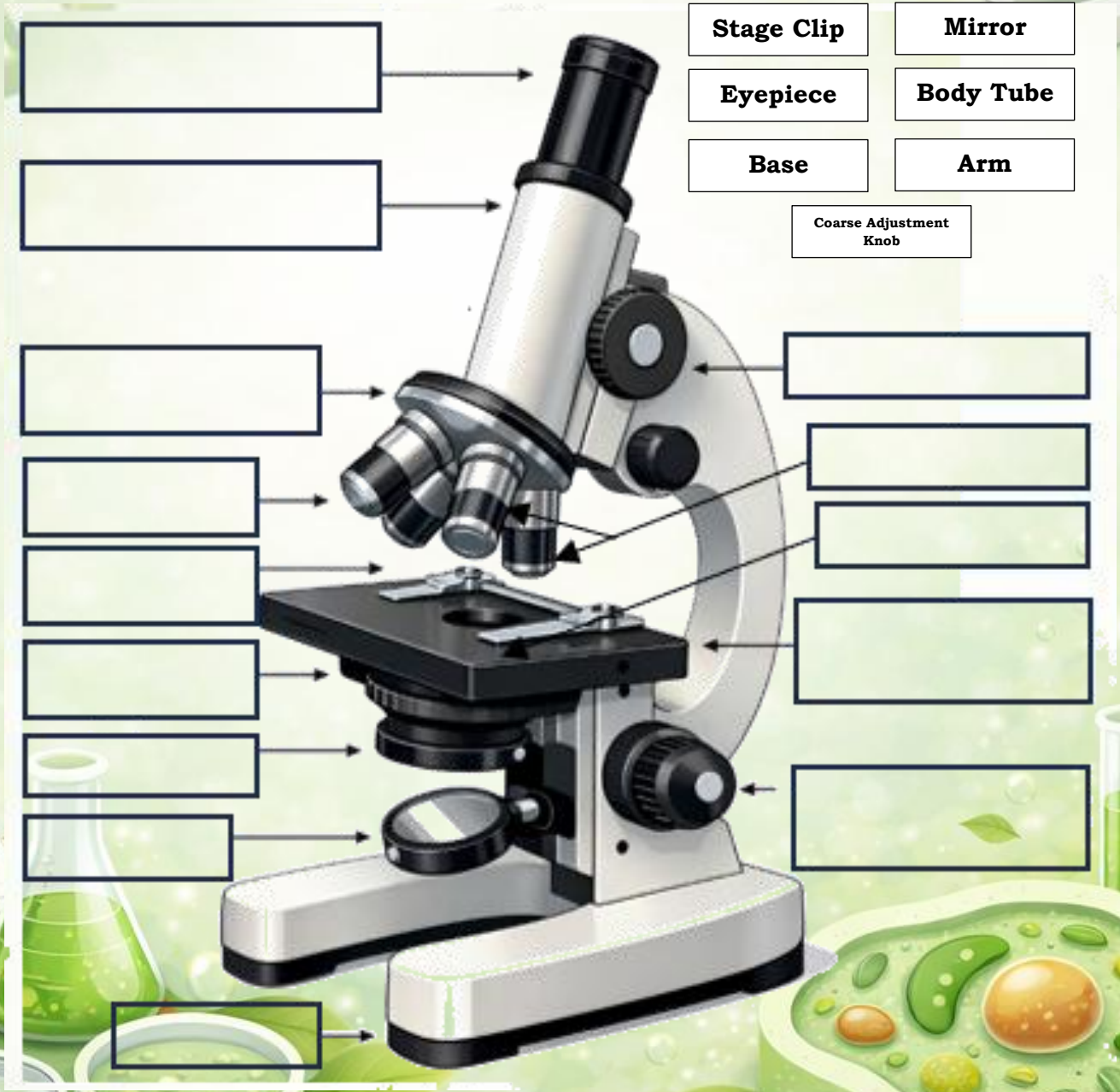
1. describe the structure and function of the light and electron microscopes;
2. compare the magnification and resolution of light and electron microscopes;
3. explain the advantages and limitations of each microscope in cell observation; and
4. analyze information from secondary sources to support explanations about microscope use in cell study.

What's In

Activity 1.1: "Zoom in! Identify the Microscope Parts"

Directions: Identify the parts of the light microscope. Cut the parts and paste it to their correct place.

Diaphragm	Nosepiece
Objective Lenses	Fine Adjustment Knob
Stage	Condenser
Stage Clip	Mirror
Eyepiece	Body Tube
Base	Arm
Coarse Adjustment Knob	



A **light microscope** is an instrument used to magnify small objects such as cells and microorganisms that cannot be seen with the naked eye. It uses visible light and lenses to enlarge the image of a specimen, allowing scientists and students to observe fine details.

Parts and Function of a Light Microscope

1. **Eyepiece (Ocular Lens)**
    - Located at the top of the microscope.
    - Magnifies the image (usually 10×).
    - This is where you look to see the specimen.
2. **Eyepiece (Ocular Lens)**
    - Located at the top of the microscope.
    - Magnifies the image (usually 10×).
    - This is where you look to see the specimen.
3. **Nosepiece (Revolving Nosepiece)**
    - Holds the objective lenses.
    - Rotates to change magnification.
4. **Stage**
    - Flat platform where the slide is placed.
    - Has clips to hold the slide in place.
5. **Light Source / Mirror**
    - Provides light that passes through the specimen.
    - Essential for making the specimen visible.
    -
6. **Light Source / Mirror**

- Provides light that passes through the specimen.
- Essential for making the specimen visible.

7. **Condenser**

- Focuses light onto the specimen.
- Improves image clarity.

8. **Coarse Adjustment Knob**

- Moves the stage up and down quickly.
- Used for initial focusing.

9. **Fine Adjustment Knob**

- Moves the stage slowly.
- Used for sharpening the image.

10. **Arm**

- Supports the upper part of the microscope.
- Used to carry the microscope.

11. **Base**

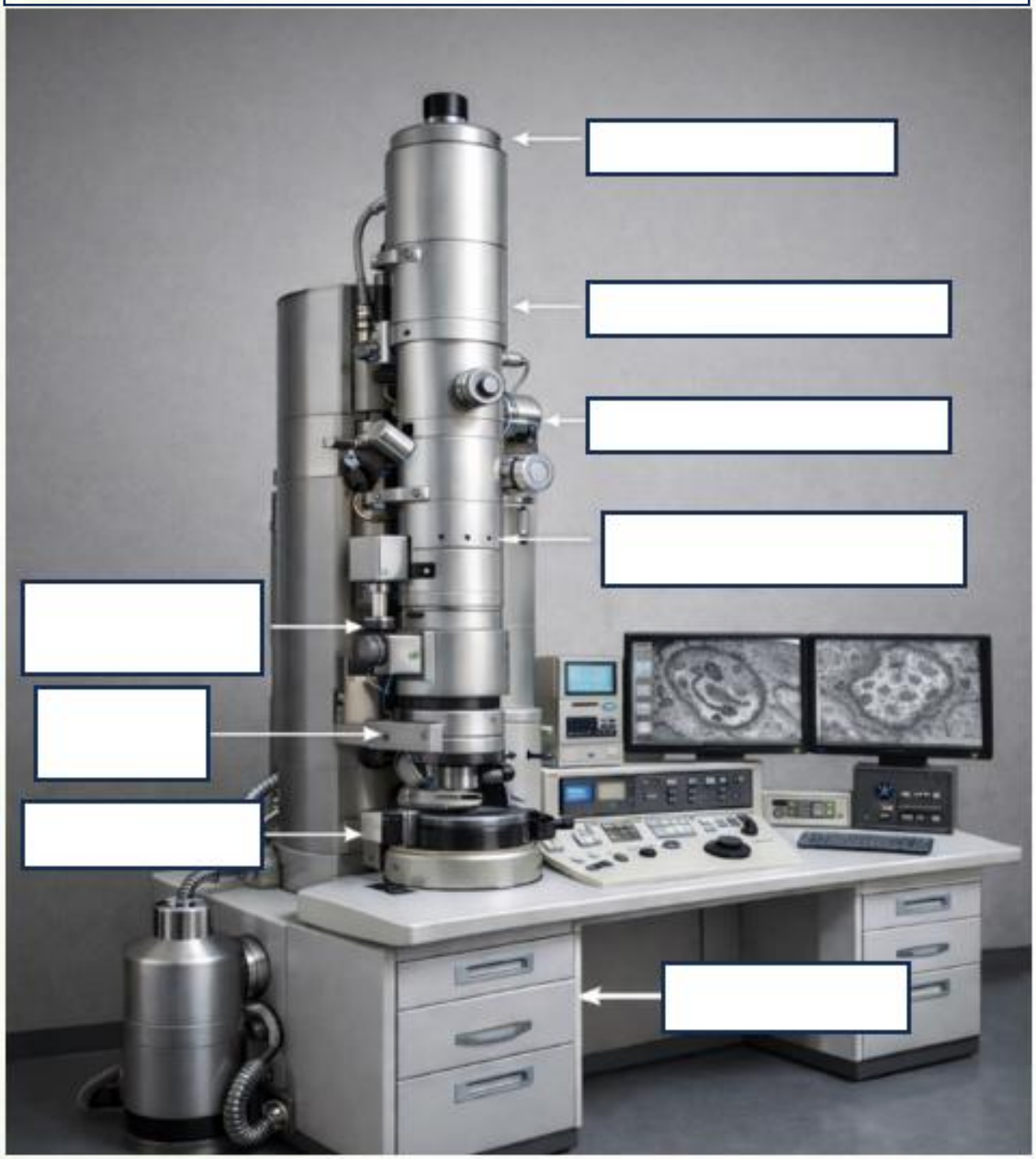
- Bottom part that supports the microscope.

- Provides stability

Activity 1.2: "Zoom in! Identify the Microscope Parts"

Directions: Identify the parts of Electron Microscope. Cut the parts and paste them to its correct place.

Specimen Holder	**Vacuum System**	**Image Viewing and Recording System**	
Electromagnetic Lenses	**Electron Gun**	**Control System**	



An **electron microscope** is a powerful scientific instrument that uses a beam of electrons instead of light to magnify extremely small structures, allowing us to see details inside cells that are invisible under a light microscope.

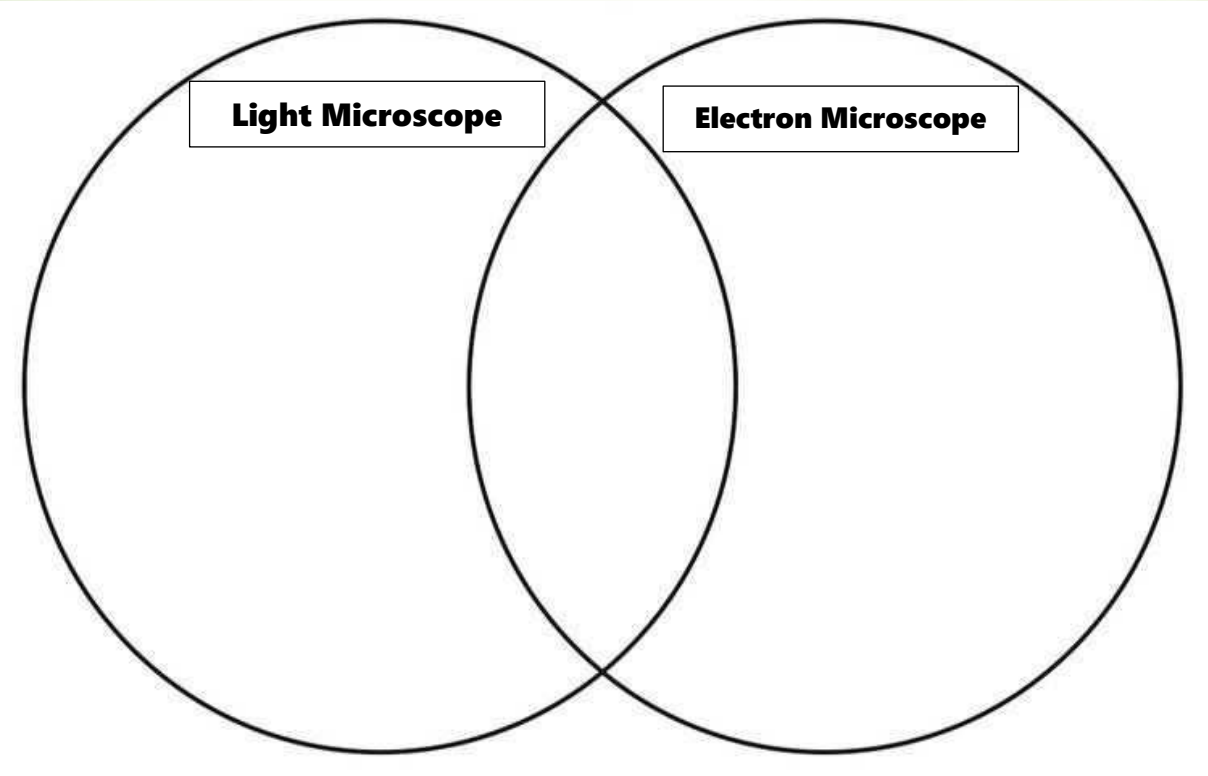
Parts and functions of Electron Microscope

1. **Electron Gun**
   - Produces a beam of electrons using a heated filament (like tungsten).
   - This replaces the light source in a light microscope.
   - **Function:** Generates the electrons needed to form an image.
2. **Electromagnetic Lenses**
   - Made of magnetic coils that control and focus the electron beam.
   - Function like glass lenses in a light microscope.
   - **Function:** Focus and direct the electron beam onto the specimen.
3. **Specimen Holder (Stage)**
   - Holds the sample in place inside the microscope.
   - The specimen must be very thin (especially in TEM).
   - **Function:** Positions the specimen for observation.
4. **Vacuum System**
   - Removes air inside the microscope.
   - Prevents electrons from scattering.
   - **Function:** Maintains a vacuum so electrons travel smoothly.
5. **Image Viewing and Recording System**
   - Converts electron signals into visible images.
   - Can be a fluorescent screen, camera, or computer monitor.
   - **Function:** Displays and records the magnified image.
6. **Control System**
   - Includes knobs, buttons, and software controls.
   - Adjusts focus, brightness, and magnification.
   - **Function:** Allows the user to operate and fine-tune the microscope.

What's New

Activity 2: "Zooming into the Invisible: A Microscope Comparison"

Directions: Compare and contrast the features, advantages, limitations, and uses of light and electron microscopes. Place its characteristics correctly and their similarity in the middle using a Venn diagram.



- Colored images
- Used in laboratories
- Very high magnification
- Can observe living cells
- Used to observe small objects
- Uses electron beam
- Very high resolution
- Produce magnified images
- Lower magnification
- Uses light
- Cannot view living cells

What's More

Activity 3: "From Colorful Cells to Invisible Structures: Tools That Bring Them to Life"

Directions: Read the passage about the light and electron microscope and answer the guide questions.

Cells are extremely small structures that cannot be seen clearly with the naked eye. Scientists use microscopes to observe cells and understand how living organism's function. Two of the most important tools used in cell study are the Light Microscope and the Electron Microscope. Each microscope has unique advantages and limitations that help scientists study cells in different ways.

The light microscope uses visible light and a series of lenses to magnify objects. It is commonly used in classrooms and laboratories because it is easy to operate and relatively inexpensive. One of the major advantages of a light microscope is that it can be used to observe living cells. This allows scientists and students to see how cells move, grow, and divide in real time. Light microscopes can magnify objects up to about 1000 times, which is enough to observe the basic structures of a cell, such as the nucleus, cell membrane, and cytoplasm.

However, the light microscope also has limitations. Because it relies on light to magnify objects, its resolution is limited. Very tiny structures inside the cell, such as viruses or detailed parts of organelles, cannot be seen clearly using this microscope. As a result, scientists sometimes need a more powerful tool to observe smaller cellular components.

The electron microscope was developed to overcome these limitations. Instead of using light, it uses a beam of electrons to create an image. Because electrons have much shorter wavelengths than visible light, electron microscopes can produce images with extremely high magnification and resolution. This allows scientists to see very small structures inside cells, including organelles, membranes, and even viruses.

Despite its powerful magnification, the electron microscope also has some disadvantages. It is very expensive and requires specialized training to operate. In addition, specimens must be placed in a vacuum and prepared using special chemicals. Because of this process, living cells cannot be observed with an electron microscope. The images produced are usually black and white, although they may be colored later for easier interpretation.

Both microscopes play an important role in the study of cells. The light microscope is useful for observing living cells and studying their general structure and behavior. On the other hand, the electron microscope allows scientists to explore the detailed internal structures of cells that cannot be seen with light microscopes. By using both tools, scientists gain a deeper understanding of the complex world inside every cell.

Guide Questions:

1. What are the main differences between a light microscope and an electron microscope? _____
_____
_____

2. Why is the light microscope useful for observing living cells?
_____
_____
_____

3. Explain one limitation of the light microscope.
_____
_____
_____

4. Why do scientists still use electron microscopes even though they are expensive?
_____
_____
_____

5. How do both microscopes help scientists understand cells better?
_____
_____
_____

To experience **MORE** interactivity in this module, scan this QR code.

What I've Learned

Directions: Encircle the letter of the correct answer.

1. A biologist wants to observe the movement of living cells in real time. Which microscope should they choose and why?
   a. Electron microscope, because it produces high-resolution images
   b. Light microscope, because it allows observation of living cells
   c. Electron microscope, because it uses electrons instead of light
   d. Light microscope, because it can observe viruses
2. A researcher studies the detailed structure of a virus inside a host cell. Which factor makes the electron microscope more suitable than the light microscope?
   a. Electron microscopes are portable and inexpensive
   b. Electron microscopes produce images with much higher magnification and resolution
   c. Electron microscopes allow real-time observation of living cells
   d. Electron microscopes use visible light for easy viewing
3. Two microscopes are available in a lab: one light microscope and one electron microscope. Which scenario demonstrates the best use of both microscopes?
   a. Observing the shape of living cells with a light microscope and examining organelle details with an electron microscope
   b. Observing viruses with a light microscope and real-time cell division with an electron microscope
   c. Using both microscopes to study only non-living specimens
   d. Observing cells with a light microscope and coloring electron microscope images
4. Which statement best explains why electron microscopes cannot observe living cells?
   a. Electron beams are too weak to pass through cells
   b. Specimens must be placed in a vacuum and treated with chemicals
   c. They produce colored images only
   d. They have lower magnification than light microscopes
5. A student argues that the light microscope is "better" than the electron microscope because it is cheaper. Which counter-argument best evaluates this claim?
   a. The cost doesn't matter if only living cells are studied
   b. Light microscopes cannot see very small structures, limiting detailed cell study
   c. Electron microscopes are more expensive but produce blurry images
   d. Light microscopes are not actually cheaper than electron microscopesGeneral